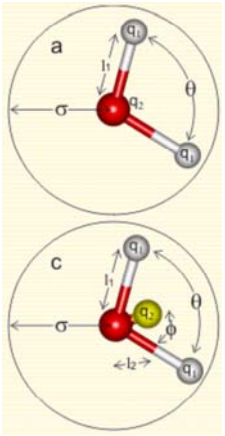
통계역학

제3장 분자동역학 모의실험 실습

중요 학습 내용

- VMD 및 NAMD 사용법
- 물분자에 대한 분자동역학 모의실험
- 물상자에서 방사분포함수 구하기
- **NaCl** 용액에서 방사분포함수 구하기

물분자 모델과 분자동역학 모의실험



parameters	SPC (original)	SPC/E (original)	TIP3P (original)	TIP3P (CHARMM)	TIP4P (original)	TIP4P (2005)
dipole (debye)[a]	2.274	2.351	2.351	2.347	2.18	2.305
$r_{\min}^{OO}$ (Å)	3.5533	3.5533	3.5365	3.5365	3.5389	3.5457
ϵ^{OO} (kcal mol^{-1})	0.1553	0.1553	0.1521	0.1521	0.1549	0.1852
$r_{\min}^{HH}$ (Å)	0	0	0	0.449	0	0
ϵ^{HH} (kcal mol^{-1})	0	0	0	0.046	0	0
$r_{\min}^{OH}$ (Å)	0	0	0	1.993	0	0
ϵ^{OH} (kcal mol^{-1})	0	0	0	0.084	0	0
q^{O} ($\|e\|$)	−0.82	−0.8476	−0.834	−0.834	0	0
q^{H} ($\|e\|$)	0.41	0.4238	0.417	0.417	0.52	0.5564
q^{M} ($\|e\|$)					−1.04	−1.1128
s_0^{OH} (Å)	1.0	1.0	0.9572	0.9572	0.9572	0.9572
s_0^{OH} (Å)					0.15	0.1546
θ_0^{HOH} (deg)	109.47	109.47	104.52	104.52	104.52	104.52
K_b (kcal mol^{-1} Å^{-2})				450.0		
K_θ (kcal mol^{-1} rad^{-2})				55.0		

[a] The experimental value for the isolated water molecule is 1.85 D.

$$V^{\mathrm{H_2O}} = \sum_i K_b(r_i - r_{i,0})^2 + \sum_i K_\theta(\theta_i - \theta_{i,0})^2$$
$$+ \sum_{i=1}^{N}\sum_{j>i}^{N} \frac{1}{4\pi\epsilon_0}\frac{q_i q_j}{r_{ij}} + \sum_{i=1}^{N}\sum_{j>i}^{N} \epsilon\left[\left(\frac{r_{\min}^{ij}}{r_{ij}}\right)^{12} - \left(\frac{2r_{\min}^{ij}}{r_{ij}}\right)^{6}\right]$$

NAMD의 input 파일 구성

- PDB 파일: 모의실험 계의 분자들의 좌표를 담고 있는 파일
- PSF 파일: 모의실험 계의 분자들에 관한 좌표 이외의 정보들 (부분전하, 질량 등)을 담고 있는 파일
- CONF 파일: NAMD를 실행하기 위한 환경 (모의실험 상자 크기, 온도, 압력 등)을 정의하는 파일

VMD를 이용해 물 상자 만들기

1) Terminal에서 "vmd"를 입력하여 VMD 띄우기
2) VMD main 창에서, "**Extensions > Modeling > Add Solvation Box**" 선택
3) "Solvate" 창에서 다음을 선택
   - Waterbox Only: Checked
   - Output 이름을 "Solvate" 대신 "watbox"로 입력 (주의: watbox 앞에 파일이 출력될 디렉토리 경로를 모두 기입해야 함)
   - Boundary: 2.4 ← 물분자 사이 최소 거리 2.4 Å 유지
   - Box Size: x, y, z min =(-15.0, -15.0, -15.0) and x, y, z max = (15.0, 15.0, 15.0)
   - Box Padding: 0 Å in all six directions
4) "Solvate" 창 맨 밑의 "**Solvate**" 버튼 클릭 → "watbox.psf"와 "watbox.pdb" 파일 생성 (VMD main 창에 watbox.psf 생성 확인)
5) 생성된 물 상자의 크기와 중심을 확인하기 위해, "**Extensions > Tk Console**" 선택
6) Tk Console 창에서 다음 명령어 입력

```
set all [atomselect top all]
measure minmax $all   (상자크기 확인)
measure center $all weight mass  (상자 중심 확인)
```

Conf 파일 만들기

다음 URL에서 watbox.conf 파일 다운로드: http://tinyurl.com/gmb4blz

```
#############################################################
## ADJUSTABLE PARAMETERS                                   ##
#############################################################

structure          ./watbox.psf
coordinates        ./watbox.pdb

set temperature    298.15
set pressure       1.00  :# bar
set outputname     watbox.result
set xaxis          30.0  :# box length in x direction
set yaxis          30.0  :# box length in y direction
set zaxis          30.0  :# box length in z direction
set total_ts       5000  :# 10 ps

firsttimestep      0

#############################################################
## SIMULATION PARAMETERS                                   ##
#############################################################

# Input
paraTypeCharmm      on
parameters          /opt/local/charmm_ff/charmm27_param/par_all27
_prot_lipid.inp
temperature         $temperature


# Force-Field Parameters
exclude             scaled1-4
1-4scaling          1.0
cutoff              12.0
switching           on
switchdist          10.0
pairlistdist        14.0

# Integrator Parameters
timestep            2.0  :# 2fs/step
rigidBonds          all  :# needed for 2fs steps
nonbondedFreq       1
fullElectFrequency  2
stepspercycle       10

# Constant Temperature Control
langevin            on    :# do langevin dynamics
langevinDamping     1     :# damping coefficient (gamma) of 1/ps
langevinTemp        $temperature
langevinHydrogen    off   :# don't couple langevin bath to hydrogens

# Periodic Boundary Conditions
cellBasisVector1    $xaxis   0.0   0.0
cellBasisVector2    0.0  $yaxis   0.0
cellBasisVector3    0.0   0.0 $zaxis
cellOrigin          0.0   0.0   0.0
```



```
wrapAll             on

# PME (for full-system periodic electrostatics)
PME                 yes
PMEGridSpacing      1.0

#manual grid definition
#PMEGridSizeX        45
#PMEGridSizeY        45
#PMEGridSizeZ        48


# Constant Pressure Control (variable volume)
useGroupPressure      yes :# needed for rigidBonds
useFlexibleCell       no
useConstantArea       no

langevinPiston        on
langevinPistonTarget  $pressure
langevinPistonPeriod  100.0
langevinPistonDecay   50.0
langevinPistonTemp    $temperature


# Output
outputName          $outputname

restartfreq         500     :# 500steps = every 1ps
dcdfreq             250
xstFreq             250
outputEnergies      100
outputPressure      100


#############################################################
############
## EXECUTION SCRIPT                                        ##
#############################################################
############
# Minimization
minimize            100
reinitvels          $temperature

run $total_ts
run 50000 :# 100ps
```


How to run NAMD

- Using a single core in a single node

  namd2 conf_file_name > log_file_name &

- Using multi cores in a single node (w/o CUDA)

  namd2 +p4 conf_file_name > log_file_name &

- Using multi cores in a single node with GPU (w/ CUDA)

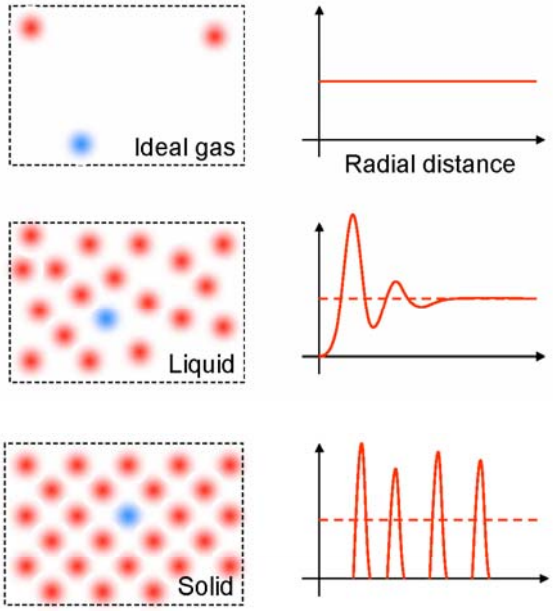
  namd2 +idlepoll +p4 conf_file_name > log_file_name &

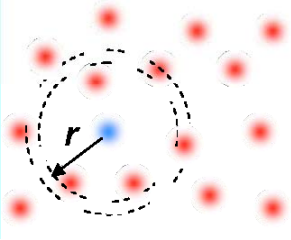
- 예제:

  namd2 +idlepoll +p4 watbox.conf > watbox.log &

방사분포함수 (Radial distribution function, RDF) - structured fluids

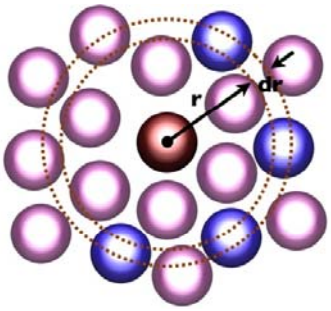
- Radial distribution function of gas, liquid, and solid

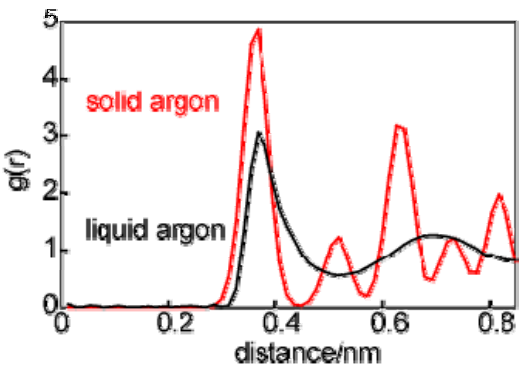


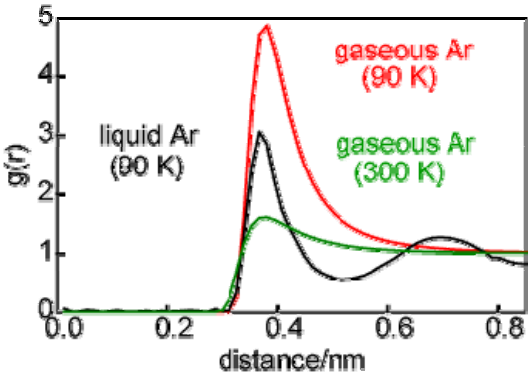


- Calculate the number of particles inside a radial segment ***(r,r+Δr)*** from one specific particle in the system.
- Average the distribution over all particles in the system.
- Divide the (average) number of particles in a segment by the segment volume.
- Finally, normalize thus obtained radial density profile with the bulk density ***N/V*** of the system.

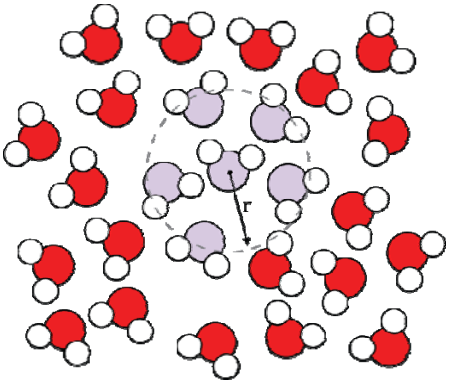
RDF of argon

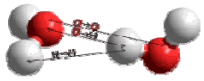


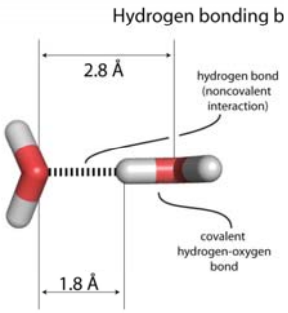


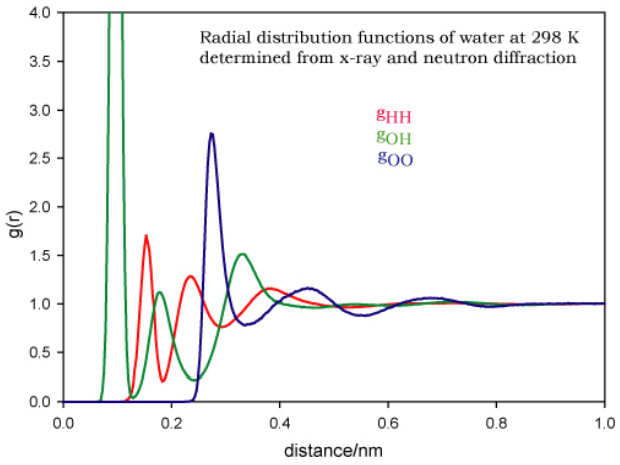


중성자 산란 실험으로부터 얻어진 물의 RDF









VMD를 이용해 물분자의 RDF 구하기

1) VMD main 창에서, "**Extensions > Analysis > Radial Pair Distribution Function g(r)**" 선택
2) "Radial Pair Distribution Function g(r)" 창에서 다음을 선택
   - Use Molecule에서 "0:watbox.psf" 선택
   - $g_{HH}(r)$ 그래프 그리기: Selection 1에 "name H1", Selection 2에 "name H1" 기입
   - $g_{OH}(r)$ 그래프 그리기: Selection 1에 "name H1", Selection 2에 "name OH2" 기입
   - $g_{OO}(r)$ 그래프 그리기: Selection 1에 "name OH2", Selection 2에 "name OH2" 기입
3) "Radial Pair Distribution Function g(r)" 창 맨 밑의 "Compute g(r)" 클릭

VMD를 이용해 NaCl 용액 만들기

1) Terminal에서 "vmd"를 입력하여 VMD 띄우기
2) VMD main 창에서, "**Extensions > Modeling > Add Ions**" 선택
3) "Solvate" 창에서 다음을 선택
   - Input PSF와 PDB: 각각 "watbox.psf"와 "watbox.pdb" 선택
   - Output prefix: "Ionized" 대신 "nacl"로 입력 (주의: nacl 앞에 파일이 출력될 디렉토리 경로를 모두 기입해야 함)
   - Choose salt: "NaCl"
   - Ion placement mode: "Neutralize and set ion concentration to" 선택, 농도 "0.5 mol/L" 선택
4) "Add Ions" 창 맨 밑의 "**Autoionize**" 버튼 클릭 → "nacl.psf"와 "nacl.pdb" 파일 생성 (VMD main 창에 nacl.psf 생성 확인)

VMD를 이용해 NaCl의 RDF 구하기

1) VMD main 창에서, "**Extensions > Analysis > Radial Pair Distribution Function g(r)**" 선택
2) "Radial Pair Distribution Function g(r)" 창에서 다음을 선택
   - Use Molecule에서 "0:nacl.psf" 선택
   - $g_{na\text{-}cl}(r)$ 그래프 그리기: Selection 1에 "name SOD", Selection 2에 "name CLA" 기입
   - $g_{na\text{-}na}(r)$ 그래프 그리기: Selection 1에 "name SOD", Selection 2 에 "name CLA" 기입
   - $g_{cl\text{-}cl}(r)$ 그래프 그리기: Selection 1에 "name SOD", Selection 2에 "name CLA" 기입
3) "Radial Pair Distribution Function g(r)" 창 맨 밑의 "Compute g(r)" 클릭

Na^+, Cl^- 이온과 물분자와의 g(r)

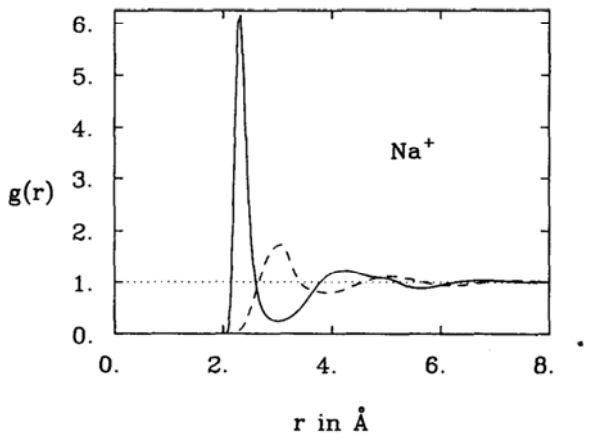


FIG. 1. Radial solute–solvent pair distribution functions for Na^+–H_2O. The solid line is $g_{NaO}(r)$. The dashed line is $g_{NaH}(r)$.

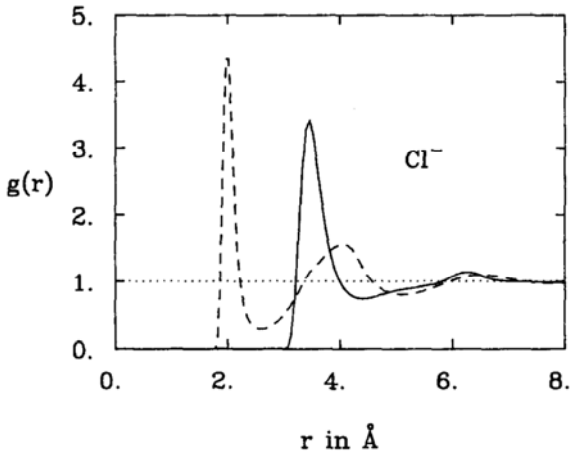
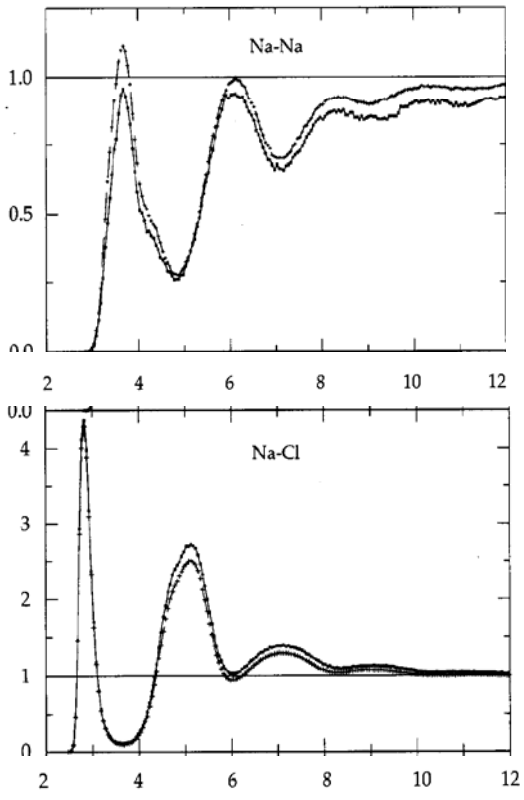


FIG. 2. Radial solute–solvent pair distribution functions for Cl^-–H_2O. The solid line is $g_{ClO}(r)$. The dashed line is $g_{ClH}(r)$.

Pettitt & Rossky, J. Chem. Phys. **84**, 5836 (2016).

Na^+-Na^+, Cl^--Cl^-, Na^+-Cl^- 의 g(r)



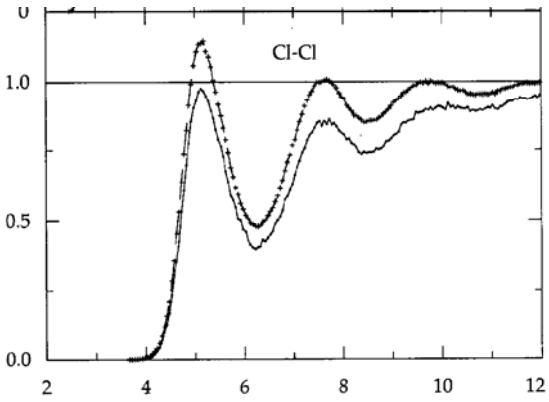


FIG. 1. Radial distribution functions for ion pairs in aqueous NaCl electrolyte. Continuous line, 0.5 M; dashed line, 1 M.

Lyubartsev & Marcelja, Phys. Rev. E, **65**, 041202 (2016).